Unit 2 - Linear Algebra
Vectors and Matrices

MSDS 520

Outline

Matrix Notation

Matrix Notation

How to write matrices:

We can assign a matrix a letter like we do with variables

- Let's call our matrix A
- We can refer to the entries of our matrix by writing the name of the matrix with subscripts to denote which row/column (respectively) we are referring to: a_{ij}, A_{ij}
- Example: a_{12} is the entry in the first row and second column
- The matrix A can also be represented by $[a_{ij}]$ (essentially means "the matrix with these entries")

$$\begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix} \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix} \begin{bmatrix} a_{11} & a_{12} & a_{13} & a_{14} \\ a_{21} & a_{22} & a_{23} & a_{24} \\ a_{31} & a_{32} & a_{33} & a_{34} \\ a_{41} & a_{42} & a_{43} & a_{44} \end{bmatrix} \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \\ a_{31} & a_{32} \end{bmatrix} \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \end{bmatrix}$$

Matrix Notation

Equality of matrices:

Suppose we have two $m \times n$ matrices A, B (note that they are the same dimension). Then $A = B$ iff each of their corresponding entries are equal, i.e., $a_{ij} = b_{ij}$ for every i, j.

$$\begin{bmatrix} 3 & -3 & 7 \\ 2 & 6 & -2 \\ 4 & 2 & 5 \end{bmatrix} = \begin{bmatrix} 3 & -3 & 7 \\ 2 & 6 & -2 \\ 4 & 2 & 5 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 4 \\ -3 & 2 \\ 4 & -7 \end{bmatrix} \neq \begin{bmatrix} 1 & 4 \\ -3 & 2 \\ 5 & -7 \end{bmatrix}$$

$$\underset{2 \times 3}{\begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix}} \neq \underset{2 \times 2}{\begin{bmatrix} 1 & 2 \\ 4 & 5 \end{bmatrix}}$$

Matrix Notation

Let a_{ij} denote a general element in A and b_{ij} a general element in B, where

$$A = \begin{bmatrix} 3 & -3 & 7 \\ 1 & 6 & -2 \\ 4 & 2 & 5 \end{bmatrix} \quad \text{and} \quad B = \begin{bmatrix} 3 & x & 7 \\ 1 & 6 & -2 \\ 4 & 5 & 2 \end{bmatrix}.$$

(a) Identify a_{12}, b_{32}, and a_{13}.
(b) Compute $a_{31} b_{13} + a_{32}b_{23} + a_{33}b_{33}$.
(c) Is there a value for x that will make the statement $A = B$ true?

Matrix Arithmetic

Matrix Arithmetic

OPERATIONS ON MATRICES

Matrix Addition

The sum of two $m \times n$ matrices A and B is the $m \times n$ matrix $A + B$, in which each element is the sum of the corresponding elements of A and B. This is written as $A + B = [a_{ij}] + [b_{ij}] = [a_{ij} + b_{ij}]$. If A and B have different dimensions, then $A + B$ is undefined.

Matrix Subtraction

The difference of two $m \times n$ matrices A and B is the $m \times n$ matrix $A - B$, in which each element is the difference of the corresponding elements of A and B. This is written as $A - B = [a_{ij}] - [b_{ij}] = [a_{ij} - b_{ij}]$. If A and B have different dimensions, then $A - B$ is undefined.

Multiplication of a Matrix by a Scalar

The product of a scalar (real number) k and an $m \times n$ matrix A is the $m \times n$ matrix kA, in which each element is k times the corresponding element of A. This is written as $kA = k[a_{ij}] = [ka_{ij}]$.

Matrix Arithmetic

Operations on matrices:

Two matrices can be added/subtracted to/from each other by adding/subtracting their corresponding entries (MUST BE THE SAME DIMENSION)

Matrix Addition: Add Corresponding Elements

$$\begin{bmatrix} 2 & 2 & 2 \\ 0 & 2 & 0 \\ 0 & 2 & 0 \end{bmatrix} + \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix} = \begin{bmatrix} 2+1 & 2+1 & 2+1 \\ 0+1 & 2+1 & 0+1 \\ 0+1 & 2+1 & 0+1 \end{bmatrix} = \begin{bmatrix} 3 & 3 & 3 \\ 1 & 3 & 1 \\ 1 & 3 & 1 \end{bmatrix}$$

Matrix Subtraction: Subtract Corresponding Elements

$$\begin{bmatrix} 3 & 3 & 3 \\ 1 & 3 & 1 \\ 1 & 3 & 1 \end{bmatrix} - \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix} = \begin{bmatrix} 3-1 & 3-1 & 3-1 \\ 1-1 & 3-1 & 1-1 \\ 1-1 & 3-1 & 1-1 \end{bmatrix} = \begin{bmatrix} 2 & 2 & 2 \\ 0 & 2 & 0 \\ 0 & 2 & 0 \end{bmatrix}$$

Matrix Arithmetic

If $A = \begin{bmatrix} 7 & 8 & -1 \\ 0 & -1 & 6 \end{bmatrix}$ and $B = \begin{bmatrix} 5 & -2 & 10 \\ -3 & 2 & 4 \end{bmatrix}$, find the following.

(a) $A + B$ **(b)** $B + A$ **(c)** $A - B$

SOLUTION

(a)
$$A + B = \begin{bmatrix} 7 & 8 & -1 \\ 0 & -1 & 6 \end{bmatrix} + \begin{bmatrix} 5 & -2 & 10 \\ -3 & 2 & 4 \end{bmatrix}$$
$$= \begin{bmatrix} 7 + 5 & 8 + (-2) & -1 + 10 \\ 0 + (-3) & -1 + 2 & 6 + 4 \end{bmatrix}$$
$$= \begin{bmatrix} 12 & 6 & 9 \\ -3 & 1 & 10 \end{bmatrix}$$

(b)
$$B + A = \begin{bmatrix} 5 & -2 & 10 \\ -3 & 2 & 4 \end{bmatrix} + \begin{bmatrix} 7 & 8 & -1 \\ 0 & -1 & 6 \end{bmatrix}$$
$$= \begin{bmatrix} 5 + 7 & -2 + 8 & 10 + (-1) \\ -3 + 0 & 2 + (-1) & 4 + 6 \end{bmatrix}$$
$$= \begin{bmatrix} 12 & 6 & 9 \\ -3 & 1 & 10 \end{bmatrix}$$

(c)
$$A - B = \begin{bmatrix} 7 & 8 & -1 \\ 0 & -1 & 6 \end{bmatrix} - \begin{bmatrix} 5 & -2 & 10 \\ -3 & 2 & 4 \end{bmatrix}$$
$$= \begin{bmatrix} 7 - 5 & 8 - (-2) & -1 - 10 \\ 0 - (-3) & -1 - 2 & 6 - 4 \end{bmatrix}$$
$$= \begin{bmatrix} 2 & 10 & -11 \\ 3 & -3 & 2 \end{bmatrix}$$

Matrix Arithmetic

Multiplication of a Matrix by a Scalar

The product of a scalar (real number) k and an $m \times n$ matrix A is the $m \times n$ matrix kA, in which each element is k times the corresponding element of A. This is written as $kA = k[a_{ij}] = [ka_{ij}]$.

$$B = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}$$

Scalar Multiplication: Multiply Each Element by the Scalar

$$2B = 2\begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix} = \begin{bmatrix} 2(1) & 2(1) & 2(1) \\ 2(1) & 2(1) & 2(1) \\ 2(1) & 2(1) & 2(1) \end{bmatrix} = \begin{bmatrix} 2 & 2 & 2 \\ 2 & 2 & 2 \\ 2 & 2 & 2 \end{bmatrix}$$

Matrix Arithmetic

Two points about sums/differences:

Like normal addition/subtraction:

- Matrix addition IS commutative ($A + B = B + A$)
- Matrix subtraction IS anti-commutative ($A - B = -(B - A)$)

$$A = \begin{bmatrix} 4 & -2 \\ 3 & 5 \end{bmatrix}, B = \begin{bmatrix} 0 & 1 \\ -2 & 3 \end{bmatrix}, C = \begin{bmatrix} 1 & -1 \\ 0 & 7 \\ -4 & 2 \end{bmatrix}, \text{and } D = \begin{bmatrix} -1 & -3 \\ 9 & -7 \\ 1 & 8 \end{bmatrix}$$

(a) $A + 3B$ **(b)** $A - C$ **(c)** $-2C - 3D$

Matrix Products

Matrix Products

MATRIX MULTIPLICATION

The **product** of an $m \times n$ matrix A and an $n \times k$ matrix B is the $m \times k$ matrix AB, which is computed as follows. To find the element of AB in the ith row and jth column, multiply each element in the ith row of A by the corresponding element in the jth column of B. The sum of these products will give the element in row i, column j of AB.

Matrix multiplication:

$$A = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix} \quad B = \begin{bmatrix} b_{11} & b_{12} \\ b_{21} & b_{22} \end{bmatrix} \quad C = \begin{bmatrix} c_{11} & c_{12} \end{bmatrix}$$

$$AB = \begin{bmatrix} (a_{11}b_{11} + a_{12}b_{21}) & (a_{11}b_{12} + a_{12}b_{22}) \\ (a_{21}b_{11} + a_{22}b_{21}) & (a_{21}b_{21} + a_{22}b_{22}) \end{bmatrix}$$

AC, BC do not exist, wrong dimensions

Matrix Products

Credits Taken

	College A	College B
Student 1	10	7
Student 2	11	4

$$A = \begin{bmatrix} 10 & 7 \\ 11 & 4 \end{bmatrix}$$

Credit Cost

	Cost per Credit
College A	\$60
College B	\$80

$$B = \begin{bmatrix} 60 \\ 80 \end{bmatrix}$$

Matrix Multiplication

$$AB = \begin{bmatrix} 10 & 7 \\ 11 & 4 \end{bmatrix}\begin{bmatrix} 60 \\ 80 \end{bmatrix} = \begin{bmatrix} 10(60) + 7(80) \\ 11(60) + 4(80) \end{bmatrix} = \begin{bmatrix} 1160 \\ 980 \end{bmatrix}$$

Matrix Products

Two points about products:

- Matrix multiplication is NOT commutative ($AB \neq BA$ in general)
- Matrix multiplication pneumonic...

If possible, compute each product using

$$A = \begin{bmatrix} 1 & -1 \\ 0 & 3 \\ 4 & -2 \end{bmatrix}, B = \begin{bmatrix} -1 \\ -2 \end{bmatrix}, C = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix}, \text{ and } D = \begin{bmatrix} 1 & -1 & 2 \\ 0 & 3 & -2 \\ -3 & 4 & 5 \end{bmatrix}$$

(a) AB **(b)** CA **(c)** DC **(d)** CD

Matrix Products

PROPERTIES OF MATRICES

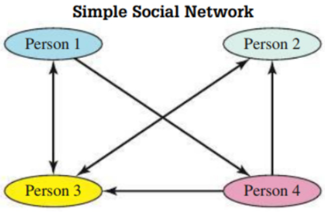
Let A, B, and C be matrices. Assume that each matrix operation is defined.

1. $A + B = B + A$	Commutative property for matrix addition; (No commutative property for matrix multiplication)
2. $(A + B) + C = A + (B + C)$	Associative property for matrix addition
3. $(AB)C = A(BC)$	Associative property for matrix multiplication
4. $A(B + C) = AB + AC$	Distributive property

Some Applications of Matrices

Matrices and Social Networks

A common modern application of matrices is to represent social networks



Use a matrix to represent the social network shown in **FIGURE 6.51**.

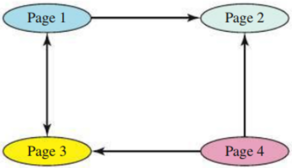
SOLUTION A social network with four people can be represented by a 4×4 square matrix. Because **Person 1** likes **Person 4**, we put a **1** in **row 1 column 4**. Similarly, **Person 4** likes **Person 2**, so we put a **1** in **row 4 column 2**. When no arrow exists to indicate that one person likes another, we place a 0 in the appropriate row and column of the matrix. Using this process results in the following matrix.

$$\begin{bmatrix} 0 & 0 & 1 & 1 \\ 0 & 0 & 1 & 0 \\ 1 & 1 & 0 & 0 \\ 0 & 1 & 1 & 0 \end{bmatrix}$$

Matrix Products

We can use matrices to represent links between webpages and how they can be navigated

Web Page Links



$$A = \begin{bmatrix} 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 1 & 0 \end{bmatrix}$$

Finding 2-click paths between web pages

Use matrix multiplication to find all 2-click paths among the web pages in **FIGURE 6.59** on the previous page.

SOLUTION

The computation A^2 can be used to determine if it is possible to get from web page i to web page j in 2 clicks (links).

$$A^2 = \begin{bmatrix} 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 1 & 0 \end{bmatrix} \cdot \begin{bmatrix} 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 1 & 0 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 1 & 1 & 0 \\ 1 & 0 & 0 & 0 \end{bmatrix}$$